

Systems of linear equations

(Sect 1.3 & 1.4 of the Notes)

Example $x + y - z = 1$
 $2x + 2y - 2z = 0$

Usually, we write such systems more concisely:

a.) as matrix times vector equation:

$$\begin{pmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

or

b.) as augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & 2 & -2 & 0 \end{array} \right)$$

Sometimes also without this line.

Example $x_1 + 2x_2 + 3x_3 = 1$
 $x_1 + x_2 + x_3 = 4$
 $2x_1 + 3x_2 + 4x_3 = 7$

corresponding matrix equation:

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix}$$

A

coefficient
matrix

x

vector
of variables

b

vector of constants

corresponding augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 1 & 1 & 1 & 4 \\ 2 & 3 & 4 & 7 \end{array} \right)$$

We want to solve such linear systems.

First question: How many solutions?

Example

$$\begin{aligned}x + y - z &= 1 \\2x + 2y - 2z &= 0\end{aligned}$$

no solution

Such a system is called **inconsistent**.

In this case, it is easy to see that there is no solution: Look at the left-hand sides of the equations. The second equation is double the first equation. However, 2 times 1 is not 0. Therefore, there are no values that you can plug in for x , y , and z that could satisfy both equations simultaneously.

Example

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 1 \\x_1 + x_2 + x_3 &= 4 \\2x_1 + 3x_2 + 4x_3 &= 7\end{aligned}$$

no solution

inconsistent

Why inconsistent?

Left-hand side: sum of first equation and second equation = third equation

Right-hand side: $1 + 4$ does not equal 7.

How can we change the system so that it has a solution?

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 1 \\x_1 + x_2 + x_3 &= 4 \\2x_1 + 3x_2 + 4x_3 &= \cancel{7}^5\end{aligned}$$

Now the system has a solution. (In fact, it even has infinitely many solutions.)

Example $x + y - z = 1$
 $2x + 2y - 2z = 2$

infinitely many solutions

indeterminate system

Why infinitely many solutions?

The second equation is just two times the first equation. Hence it is redundant.
We are left with the single equation

$$x + y - z = 1,$$

which obviously has infinitely many solutions.

Example $x_1 + x_2 + x_3 = 3$
 $x_2 + x_3 = 2$
 $x_3 = 1$

unique solution

The only solution is $x_1 = x_2 = x_3 = 1$, which you can see by backsubstitution.

Summary

There are only three possibilities:

- no solution (inconsistent system)
- exactly one solution (unique solution)
- infinitely many solutions (indeterminate system)

How to solve a linear system of equations

Easy if the system is in "triangular form":

Example $x_1 + x_2 + x_3 = -2$

$$3x_2 + x_3 = 8$$

$$2x_3 = 10$$

Backsubstitution:

$$\underline{x_3 = 5}$$

$$3x_2 + 5 = 8$$

$$\underline{x_2 = 1}$$

$$x_1 + 1 + 5 = -2$$

$$\underline{x_1 = -8}$$

The (unique) solution is therefore

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -8 \\ 1 \\ 5 \end{pmatrix}$$

In set notation: The solution set is the set $\left\{ \begin{pmatrix} -8 \\ 1 \\ 5 \end{pmatrix} \right\}$.

Still relatively easy to solve is the following:

$$x_1 + 5x_2 + 2x_3 + 3x_4 = 1$$

$$x_3 + 3x_4 = 2$$

"staircase form"

$$2x_4 = 4$$

Corresponding augmented matrix:

$$\left(\begin{array}{cccc|c} 1 & 5 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 2 & 4 \end{array} \right)$$

Pivots

Row echelon form

In this lecture, we concentrate on systems given in staircase/row echelon form.

Definition (row echelon form)

- 1.) pivot = an entry of a matrix that is the first non-zero entry in a row
- 2.) A matrix is in row echelon form if
 - rows consisting only of zeros are at the bottom of the matrix, and
 - each pivot is to the right of the pivot in the row above it

Examples

$$\begin{pmatrix} \textcircled{2} & 1 & 3 & 5 \\ 0 & \textcircled{1} & 7 & 5 \\ 0 & 0 & \textcircled{-3} & 1 \\ 0 & 0 & 0 & \textcircled{6} \end{pmatrix}$$

pivots

row echelon form

$$\begin{pmatrix} \textcircled{3} & 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 0 & \textcircled{2} & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 & 0 & \textcircled{1} & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

row echelon form

$$\begin{pmatrix} 0 & \textcircled{3} & 1 & 2 & 3 \\ 0 & 0 & \textcircled{7} & 1 & 2 \\ 0 & 0 & \textcircled{8} & 1 & 2 \\ 0 & 0 & 0 & \textcircled{4} & 1 \end{pmatrix}$$

not in row echelon form

We still want to solve the following system:

$$\left(\begin{array}{cccc|c} \textcircled{1} & 5 & 2 & 3 & 1 \\ 0 & 0 & \textcircled{1} & 3 & 2 \\ 0 & 0 & 0 & \textcircled{2} & 4 \end{array} \right) \quad \text{Row echelon form}$$

The first thing we notice is that there is at least one solution.

Why? Because there is no row of the type " $0\ 0\ 0\ 0\ 0\ 0\ |\ \neq 0$ "

To solve a linear system, we make the following main distinction:

a.) pivot columns: columns that have at least one pivot

the corresponding variables are dependent variables

b.) non-pivot columns: columns without any pivot

the corresponding variables are free variables

Hence in our example:

free variable dependent variables

x_1 x_2 x_3 x_4

$$\left(\begin{array}{cccc|c} \textcircled{1} & 5 & 2 & 3 & 1 \\ 0 & 0 & \textcircled{1} & 3 & 2 \\ 0 & 0 & 0 & \textcircled{2} & 4 \end{array} \right)$$

Free variables can be chosen arbitrarily (freely). They can take any value.

We write

$$x_2 = s, \quad s \in \mathbb{R} \text{ arbitrary}$$

Then we write how the dependent variables depend on the values of the free variables:

The last row tells us that

$$2x_4 = 4$$

Hence

$$x_4 = 2$$

This variable just happens not to depend on any free variable.

The second row tells us that

$$x_3 + 3x_4 = 2$$

Hence

$$x_3 = -4$$

This variable just happens not to depend on any free variable.

The first equation tells us that

$$x_1 + 5x_2 + 2x_3 + 3x_4 = 1$$

hence

$$x_1 + 5s + 2 \cdot (-4) + 3 \cdot 2 = 1$$

which implies

$$x_1 = 3 - 5s$$

This equation tells us how x_1 depends on s .

Therefore, we can write the solution as

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 3 - 5s \\ s \\ -4 \\ 2 \end{pmatrix} = s \begin{pmatrix} -5 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ -4 \\ 2 \end{pmatrix}$$


In set notation:

$$\left\{ s \begin{pmatrix} -5 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ -4 \\ 2 \end{pmatrix} : s \in \mathbb{R} \right\}$$

solution set

Given a system in row echelon form, there are the following steps you can take to identify its solution set.

1.) determine if the system is inconsistent.

The system is inconsistent if and only if there is a row of the type

$$0000000 \mid \neq 0$$

such as for example $0000 \mid 5$

2.) identify the free variables (those corresponding to columns without pivots)

Assign them as parameters s, t, u , etc. (or $s_1, s_2, s_3, \dots$), which can assume any real value independently of each other.

3.) identify the dependent variables (those corresponding to pivot columns)

Write how they depend on the values of the free variables.

Given a system in row echelon form, you can easily determine the number of solutions as follows:

First, determine if the system is inconsistent (see previous page).

If the system is not inconsistent
then

a.) every column has a pivot
—————→ unique solution

b.) not every column has a pivot
—————→ infinitely many solution (because there exist free variables)

Homogeneous systems

A homogeneous system is a system where all entries on the right-hand side are equal 0.

Example

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{array} \right)$$

Such a system always has at least one solution: The solution where all variables are equal to 0. Such a solution is called "trivial solution".